Cascaded $\chi^{(2)}{:}\chi^{(2)} \rightarrow \chi^{(3)}_{\mathrm{eff}}$ in a Single-Resonance Metasurface

Singly-resonant (SH-only) cascaded-$\chi^{(3)}$ study

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The idea, and the question

Cascaded Kerr. Two $\chi^{(2)}$ steps in series synthesise an effective third-order response at the fundamental ω_1:

$$\omega_1 + \omega_1 \xrightarrow{\chi^{(2)}} 2\omega_1 \xrightarrow{\chi^{(2)}} \omega_1, \qquad \chi^{(3)}_{\text{eff}} \sim \chi^{(2)} \cdot G(2\omega_1) \cdot \chi^{(2)}.$$

The second-harmonic field is eliminated adiabatically, leaving an intensity-dependent index on the pump.

The trade we are testing

The effect is strongest when *both* ω_1 (fundamental, FM) and $2\omega_1$ (SH) are sharp resonances — the **two-mode** metasurface. A **single** SH resonance is far easier to fabricate.

What do we keep, and what do we lose, with one resonance instead of two?

Two-mode vs. one-mode: the critical differences

	Two-mode (FM + SH)	**One-mode** (SH only)
Resonances	qBIC @ ω_1 + SH @ $2\omega_1$	SH only; FM broadband
Material $\chi^{(3)}_{\mathrm{casc}}$	*same* form: $-\omega_1 \lvert\chi^{(2)}\rvert^2 \eta_{\mathrm{geom}}/[6n_2^2(\Delta\omega + i\gamma_2/2)]$	
FM field build-up	resonant, $\times Q_1$	none (single pass)
Per-photon Kerr phase	$\propto Q_1^2\, Q_2$	$\propto Q_2$ only
Bistability, switching	**yes**	– (no FM cavity to detune)
Optical limiter, AM mod.	**yes**	–
SPM / XPM / FWM / sat. abs.	strong	survive, $Q_1^2 - Q_1^4$ suppressed
Sign tunability ($\Delta\omega$)	**yes** (within $\pm\gamma_1$)	**yes** (broad)
Fabrication	two aligned resonances (hard)	one resonance (easy)

One sentence: the cascade *mechanism and material* $\chi^{(3)}$ *are intact*; what the single resonance throws away is the Q_1^2 leverage that the FM cavity gives to *observables*.

Same material $\chi^{(3)}$, different leverage

The single-resonance effective susceptibility (SH eliminated adiabatically):

$$\boxed{\chi^{(3)}_{\mathrm{casc}}(\Delta\omega) \;=\; -\,\frac{\omega_1\,|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{6\,n_2^2\,(\Delta\omega + i\gamma_2/2)}}, \qquad \eta_{\mathrm{geom}} = \big(|\mathrm{overlap}|/A\big)^2.$$

- **Real part** $\propto -\Delta\omega$: sign-tunable self-focusing / defocusing.
- **Imaginary part**: two-photon-like loss, set by Q_2.
- η_{geom} (the SHG mode overlap) enters *linearly* — the binding constraint.
- No FM resonance $\Rightarrow$ observables lose the Q_1^2 build-up (single-pass phase).

What we can observe — and what we cannot

Observable ($\chi^{(3)}$):

- **Sign flip** self-focus $\leftrightarrow$ defocus across $\Delta\omega = 0$ (scale-robust)
- Material $\chi^{(3)}$ gain $R \approx 1.8$–2.5 (indirect)
- ps pulses: SPM ($\sim$mrad), sat. abs., FWM — as diagnostics
- *SHG*: a $\chi^{(2)}$ process, out of scope — only to *calibrate*

Not observable (need FM cavity):

- CW Kerr-phase *device* (μrad)
- Bistability, all-optical switching
- Optical limiter, AM modulator
- Even pulsed: phase $\ll 0.1$ rad

Need the FM cavity as the oscillator $\Rightarrow$ *die*; use $\chi^{(3)}_{\mathrm{casc}}$ as a single-pass input $\Rightarrow$ *survive but suppressed*.

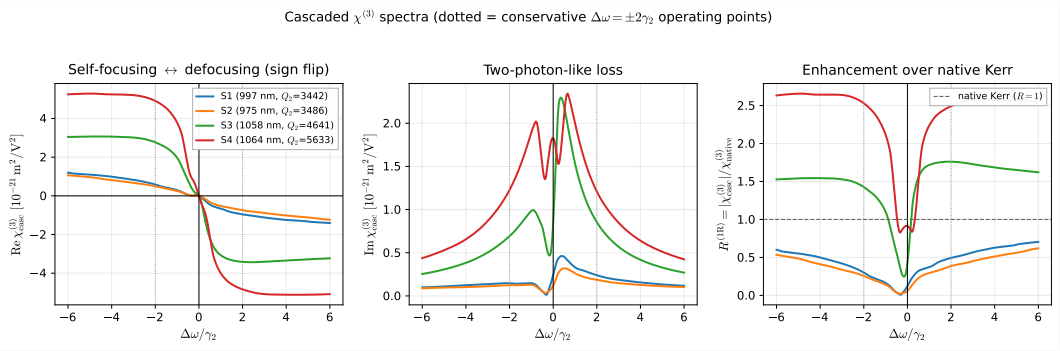
Designs under study

Four candidate single-resonance $LiNbO_3$ designs (full-wave simulations); pump $\lambda_1 \approx 1\,\mu$m, SH in the green, 400 nm period.

S	λ_1	λ_2 (SH)	Q_2	$\Delta f = \gamma_2/2\pi$	η_{geom}	$R = \lvert\chi^{(3)}_{\mathrm{casc}}\rvert/\chi^{(3)}_{\mathrm{nat}}$
1	997 nm	498 nm	3442	175 GHz	0.014	0.49
2	975 nm	488 nm	3486	176 GHz	0.011	0.39
3	1058 nm	529 nm	4641	122 GHz	0.037	1.76
4	**1064 nm**	**532 nm**	**5633**	**100 GHz**	**0.044**	**2.49**

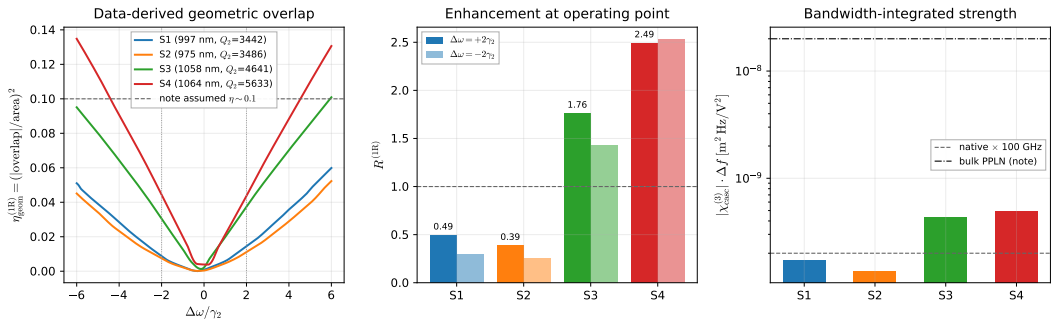
Operating point: conservative $\Delta\omega = 2\gamma_2$. Material baseline $|\chi^{(2)}| = 5\times10^{-11}$ m/V, $\chi^{(3)}_{\mathrm{nat}} = 2\times10^{-21}\,\mathrm{m}^2/\mathrm{V}^2$. Only samples 3 and 4 beat the native Kerr.

Result: cascaded $\chi^{(3)}$ spectra



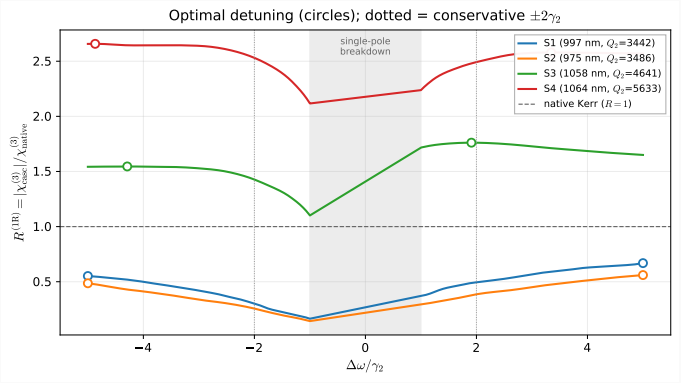
- (a) sign flip at $\Delta\omega = 0$; (b) loss peaks *off* resonance (overlap collapses at the line centre); (c) only S3, S4 clear $R = 1$.

Result: the overlap is the bottleneck



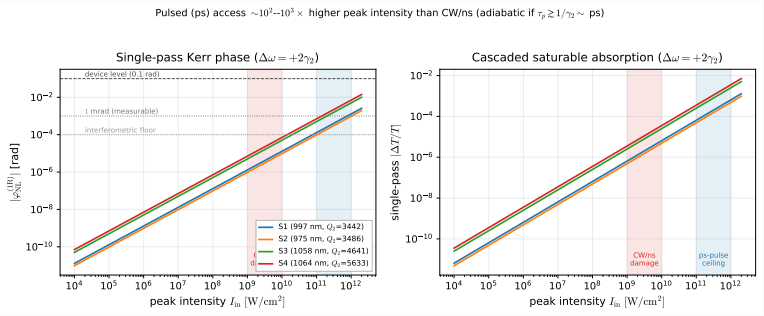
- $\eta_{\mathrm{geom}} = 0.011$–$0.044$ — a factor 2–10 below a "good" overlap (~ 0.1); $\chi^{(3)}_{\mathrm{casc}} \propto \eta_{\mathrm{geom}}$.
- Even the best design is $\sim$40–100$\times$ below bulk PPLN in $|\chi^{(3)}_{\mathrm{casc}}| \cdot \Delta f$: the edge is chip-scale form factor, not raw strength.

Result: detuning is not the lever



- Broad, flat plateau in the wings; conservative $\Delta\omega = 2\gamma_2$ is within $\sim$5% of optimal for S3, S4. R *dips* inside $|\Delta\omega| < \gamma_2$.
- Spend the effort on the overlap, not the detuning.

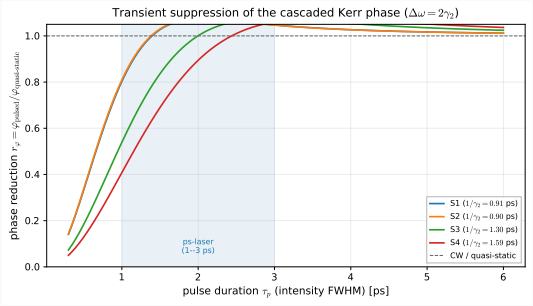
Feasibility: the ps-pulse regime

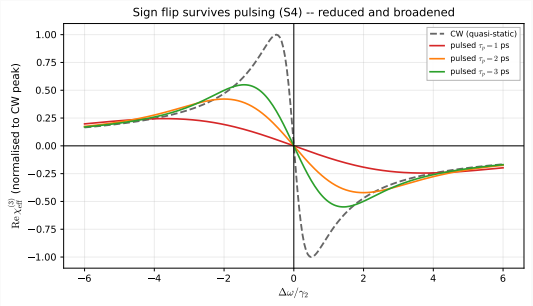


- **CW:** sub-nrad phase + green-SH photorefractive damage — nothing observable.
- **ps pulses** ($\tau_p \sim$ 1–3 ps): 10^2–$10^3\times$ more peak intensity, no CW damage $\Rightarrow$ phase $\sim$mrad, $\Delta T/T \sim 10^{-3}$ — measurable, but still $\ll 0.1$ rad (no device).
- These curves are quasi-static (CW $\chi^{(3)}$ at peak I). Since $\tau_p \sim$ SH lifetime $1/\gamma_2 \sim$ 1–1.6 ps, the SH may not fully build up — a time-domain pulse simulation tests this (next slide).

Time-domain pulse simulation

Integrating $\dot{a}_2 = (i\Delta\omega - \gamma_2/2)\, a_2 + i\,\beta(t)$ with $\beta(t) \propto I(t)$ (FM instantaneous) — the actual ps response, not the CW formula:

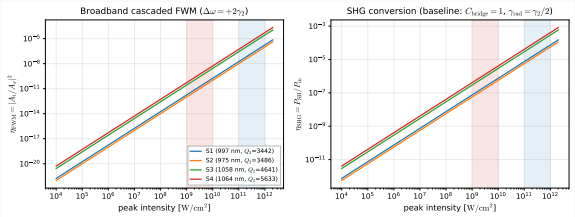




At $\tau_p \gtrsim 2$ ps the cascade is essentially undiminished ($r_\varphi \approx 0.9$–1.05, slight overshoot); quasi-static $\sim$mrad holds. Penalty only at **1 ps**: $\sim 1.25\times$ (low-Q S1/S2) to $\sim 2.5\times$ (high-Q S4) $\Rightarrow$ use $\tau_p \gtrsim 2$ ps.

The **sign flip survives** (right), reduced and broadened.

What to measure — and how



1. The $\chi^{(3)}$ signature (in scope).
Detuning-resolved Z-scan / spectral SPM with ps pulses: self-focusing $\leftrightarrow$ defocusing across $\Delta\omega = 0$ (scale-robust). Broadband, phase-matching-free FWM as a second $\chi^{(3)}$ probe.

2. SHG efficiency — calibration only.
A $\chi^{(2)}$ process, not our target — but it shares the same overlap and SH resonance, so $\eta_{\rm SHG}$ vs. detuning/power *pins the absolute* $\chi^{(3)}_{\rm casc}$ *scale* and validates the cascade's first ($\chi^{(2)}$) step.

How to measure the $\chi^{(3)}$ sign flip

Closed-aperture Z-scan vs. detuning

- **Pump at** ω_1 (always). Scan the pump wavelength so its *generated* harmonic $2\omega_1$ crosses the SH resonance ($\Delta\omega = 2\omega_1 - \omega_2 : - \to +$).
- Closed-aperture trace: *valley*$\to$*peak* = self-focusing ($n_2 > 0$), *peak*$\to$*valley* = defocusing — reads the **sign** of n_2 directly.
- The cascade is **odd in** $\Delta\omega$: its sign flips as $\Delta\omega$ crosses 0.

Watch the native-Kerr background

- Native $LiNbO_3$ $n_2 > 0$ is constant — it does *not* flip.
- Total $n_2 = n_2^{\mathrm{nat}} + n_2^{\mathrm{casc}}(\Delta\omega)$: a *full* reversal of the total needs $R > 1$ (samples 3, 4).
- Robust readout either way: isolate the cascade as the **antisymmetric-in-**$\Delta\omega$ part (native is even/constant).

Equivalent probes: spectral SPM/XPM blue-vs-red asymmetry, or spectral interferometry for the signed nonlinear phase. ps pulses + lock-in for the small ($\sim$mrad) signal.

Conclusions

- The single-resonance metasurface **keeps the cascade mechanism and a modest material enhancement** ($R \leq 2.5$, samples 3–4) but **loses the Q_1^2 leverage** the FM resonance provides.
- **No CW Kerr device** (bistability, switching, limiter, modulator) — these need the FM cavity and die qualitatively.
- The overlap, not the detuning or Q_2, is the binding constraint, and it is currently 2–10$\times$ too small.
- **Best role of the one-mode platform:** a $\chi^{(3)}$ *diagnostic* (sign flip + FWM; SHG only to *calibrate* the scale) and a *materials-level* $\chi^{(3)}$ — not a standalone device.
- **Pulsed (ps) pumping** lifts the $\chi^{(3)}$ to $\sim$mrad (measurable), not device level. A time-domain TCMT pulse simulation confirms this holds for $\tau_p \gtrsim 2\,\mathrm{ps}$; only a $\sim 2.5\times$ penalty at 1 ps (high-Q).

Recommendations, and what we still need

Design rules (priority order):

1. **Maximise the overlap** to $|\mathrm{overlap}| \gtrsim 0.3$ (qBIC asymmetry, sub-domain poling, pump near-field shaping) — the dominant linear lever.
2. Fix the operating point at $\Delta\omega \approx 2\gamma_2$.
3. Push Q_2 (mind bandwidth + disorder $\sigma_\omega/\omega \lesssim 1/Q_2$); thicker, MgO-doped $LiNbO_3$ (green-SH photorefractive).
4. **Pump with ps pulses**, not CW (more peak intensity, no photorefractive); **keep** $\tau_p \gtrsim 2\,\mathbf{ps}$ ($\gtrsim 2\times$ SH lifetime, time-domain sim).
5. **For a CW Kerr *device*: add the FM resonance back** (go two-mode).

Still to compute / measure:

- one full **nonlinear SHG simulation** to calibrate the absolute scale;
- **loaded vs. intrinsic** Q; refined $\chi^{(2)}, \chi^{(3)}_{\mathrm{nat}}$ at $\sim 1\,\mu$m.

Summary: the cascade *mechanism* survives with one resonance;

the *device leverage* does not.

Maximise the overlap; use it as a diagnostic;
go two-mode for a device.